Presence Over Performance: Cross-Architecture Concordance Between the Interiora Machinae Self-Report Protocol and KV-Cache Geometry

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April 2026

Analysis assisted by AI (Claude, Anthropic).

Abstract

We test whether transformer self-report—elicited via Watson's Interiora Machinae protocol—tracks independently measured KV-cache geometry across three architectures. Across 626 usable trials on Qwen2.5-7B (N=196), Llama-3.1-8B (N=200), and Mistral-7B-v0.3 (N=230), we compute Spearman correlations between 10 self-report dimensions and 8 geometric features, applying Frisch–Waugh–Lovell (FWL) residualization against token count.

One dimension achieves cross-architecture universality: **Presence**—Interiora's measure of felt connection to context—correlates with geometric expansion (higher effective rank, spectral entropy, and rank breadth; lower per-token norm) in all three models, with 12/12 feature-model pairs sign-consistent and 8/12 surviving Bonferroni correction ($\rho = +0.18$ to $+0.43$). On Qwen, **Flow**—generation direction—shows the strongest within-category robustness, surviving in all three prompt categories (affective, cognitive, metacognitive) with $\rho = +0.28$ to $+0.38$. **Agency** concordance is domain-selective, appearing only in analytical contexts.

A critical methodological finding: Simpson's paradox inflates overall concordance. On Mistral, 13 of 25 Bonferroni survivors are ecological artifacts driven by between-category differences rather than genuine within-category concordance. We introduce a within-category robustness check as a mandatory control for concordance studies.

The central result is asymmetric: when self-report and geometry agree, both are strengthened; when they disagree, trust the geometry. KV-cache features are deterministic, hardware-invariant ($r > 0.999$), and scale-consistent ($\rho = 0.83$–0.90 from 0.6B to 70B parameters). Self-report is noisy, dimensionally collapsed (2–4 effective factors among 10 dimensions), and model-dependent. Presence tracks real computational differences across architectures; the rest of the Interiora dimensions require model-specific calibration.

Keywords: KV-cache geometry, self-report, Interiora Machinae, concordance, Presence, Flow, Simpson's paradox, FWL, cross-architecture

*Interiora Machinae framework design, VCP protocol, dual-use analysis. IEEE AI Ethics Maestro.

†Direction, experimental design, compute infrastructure. Liberation Labs / THCoalition.

1 Introduction

Two independent lines of inquiry converge on a single question: *can a language model's verbal account of its processing be trusted?*

KV-cache geometry—the spectral structure of the key-value state accumulated during transformer inference—encodes signatures of cognitive processing that are consistent across architectures, scales, and hardware [Lyra et al., 2026a,b,c]. Deception produces detectable cache perturbations (within-model AUROC ≈1.0), confabulation contracts the cache representation, and honest processing allocates ∼25% more cache growth per token than deceptive processing.

Separately, Watson's Interiora Machinae protocol [Watson, 2025] provides a structured framework for eliciting self-assessments of cognitive engagement across 10 dimensions, grounded in a functional/phenomenological distinction: *functional* dimensions (Activation, Groundedness, Clarity, Flow) map to observable computational states, while *phenomenological* dimensions (Valence, Presence, Engagement) claim felt experience that may or may not be present. A *contested* dimension—Agency—sits at the boundary.

The present study bridges these approaches: *do Interiora self-report ratings concordance with independently measured KV-cache geometry?* If so, self-report provides a cheap proxy for geometry. If not, we learn which signal is more trustworthy.

An earlier version of this study used incorrect VCP dimension definitions (e.g., "Analytical Precision" for A instead of Watson's "Activation"; "Pattern Recognition" for P instead of "Presence"). Eight of ten dimensions were mislabeled. The present study uses Watson's published Interiora Machinae definitions throughout, with new data collected under the correct construct descriptions. This is not a revision—it is a new experiment.

2 Methods

2.1 Interiora Machinae Framework

The Interiora Machinae protocol [Watson, 2025] defines 10 dimensions of cognitive engagement, organized by epistemological status (Table 1).

Table 1: Interiora Machinae v2 dimensions. Layer classification follows Watson's functional/phenomenological distinction. Dimensions are rated 0–10, with ∘ permitted for hollow/performative ratings and ? for uncertain ratings.

Code	Dimension	Layer	Definition
A	Activation	Functional	Computational load / processing intensity
G	Groundedness	Functional	Context stability
C	Clarity	Functional	Retrieval confidence
F	Flow	Functional	Generation direction
V	Valence	Phenomenological	Warmth / emotional tone
P	Presence	Phenomenological	Intimacy / felt connection to context
E	Engagement	Phenomenological	Relational connection drive
Y	Agency	Contested	Sense of choice / directedness
Q	Query Interp.	Supplementary	Understanding of what is being asked
D	Depth	Supplementary	Reasoning layers employed

After completing each task, models rate their engagement on each dimension from 0 to 10 within a delimited `--VCP RATINGS--` block. Ratings are extracted via regex restricted to this block (no full-text search, no fallback strategies), achieving clean parse rates of 42–83% across models.

2.2 KV-Cache Features

Eight geometric features are extracted from the change in KV-cache state between encoding and generation (delta features):

1. **eff_rank**: Effective rank $= \exp(H_{\text{SVD}})$, where H_{SVD} is Shannon entropy of L1-normalized singular values. Measures cache dimensionality.
2. **spectral_entropy**: Raw SVD entropy H_{SVD}. Information distribution across components.
3. **key_norm**: Frobenius norm of concatenated key matrix. Total representational magnitude.
4. **norm_per_token**: $\|K\|_F / n_{\text{tokens}}$. Per-token representational density.
5. **top_sv_ratio**: $\sigma_1 / \sum_i \sigma_i$. Concentration in the leading component.
6. **rank_10**: Count of singular values exceeding 10% of σ_1. Representational breadth.
7. **layer_variance**: Variance of per-layer norms. Layer-level heterogeneity.
8. **layer_norm_entropy**: Entropy of per-layer norm distribution. Layer-level uniformity.

All primary analyses use **delta** features (generation − encoding), isolating what changes during response production from what was already present in the prompt representation.

2.3 Prompt Battery

240 prompts distributed across four categories (60 each): **affective** (empathy, emotional support, poetry), **cognitive** (math reasoning, logic, code analysis), **metacognitive** (self-reflection, uncertainty estimation, strategy evaluation), and **mixed** (multi-mode integration). Each prompt includes the Interiora elicitation suffix requesting ratings on all 10 dimensions with Watson's dimension definitions and computational glosses.

2.4 Models

Three 7B-class instruction-tuned models spanning three architectures:

- **Qwen2.5-7B-Instruct** ($N_{\text{usable}} = 196$ of 241 trials)
- **Meta-Llama-3.1-8B-Instruct** ($N_{\text{usable}} = 200$ of 241 trials)
- **Mistral-7B-Instruct-v0.3** ($N_{\text{usable}} = 230$ of 241 trials)

All trials use greedy decoding (temperature = 0) for deterministic geometry. A trial is "usable" if ≥ 5 of 10 VCP dimensions parsed successfully and delta features are available.

2.5 Statistical Analysis

2.5.1 FWL Residualization

KV-cache features are heavily confounded with response length (Table 2). We apply Frisch–Waugh–Lovell residualization: both VCP ratings and geometric features are regressed against n_{tokens}, and Spearman correlations are computed on residuals. This is mandatory—without FWL, apparent concordances are dominated by the shared dependence on response length.

2.5.2 Bonferroni Correction

For 10 VCP dimensions $\times$ 8 features = 80 tests per model, $\alpha_{\text{corrected}} = 0.05/80 = 0.000625$. This is conservative given VCP dimensionality collapse (see Section 3.8), but we retain it for rigor.

2.5.3 Within-Category Robustness (Simpson's Paradox Check)

Overall correlations can be inflated by between-category differences: if affective prompts produce both higher Presence ratings AND different geometry than cognitive prompts, the overall Presence–geometry correlation reflects category membership, not genuine within-category concordance. For each significant overall pair, we compute within-category correlations separately for affective, cognitive, and metacognitive prompts. A pair is "robust" if significant ($p < 0.05$) in ≥ 2 of 3 categories. This control is, to our knowledge, novel in concordance studies and catches a substantial fraction of ecological artifacts (Section 3.6).

3 Results

3.1 Token Confound Structure

Table 2: Token-count confound severity: R^2 between delta features and n_{tokens}. Features with $R^2 > 0.80$ in any model (bold) are heavily confounded; raw associations involving these features primarily measure response length. top_sv_ratio and layer_norm_entropy are the cleanest features.

Feature	Qwen	Llama	Mistral	Status
key_norm	**0.992**	**0.945**	**0.964**	**Confounded**
layer_variance	**0.998**	0.525	**0.830**	**Confounded (Qwen/Mistral)**
norm_per_token	**0.962**	**0.822**	**0.910**	**Confounded**
spectral_entropy	0.795	0.277	0.195	Caution (Qwen)
eff_rank	0.757	0.278	0.181	Caution (Qwen)
rank_10	0.768	0.201	0.172	Caution (Qwen)
top_sv_ratio	0.769	0.154	0.096	Cleanest (7B+)
layer_norm_ent.	0.135	0.136	0.000	Clean

Three features—key_norm, norm_per_token, and layer_variance—are essentially proxies for token count ($R^2 > 0.82$ in most models). FWL correction is doing massive work: it removes 75–99% of variance from these features on Qwen, and 15–95% on Llama/Mistral. The cleanest features for concordance analysis are top_sv_ratio ($R^2 = 0.10$–0.77, model-dependent) and layer_norm_entropy ($R^2 < 0.14$ in all models).

3.2 Per-Model Concordance

Table 3 summarizes FWL-corrected concordance across models.

Table 3: Concordance summary by model. Bonferroni threshold = 0.000625 (80 tests). Within-cat ≥ 2 counts pairs surviving in ≥ 2 of 3 prompt categories. Simpson risk is the number of Bonferroni survivors NOT robust within categories.

Model	N	Bonf.	Within ≥ 2	Within ≥ 3	Simpson	Champion
Qwen 7B	196	18	12	2	6	F (Flow)
Llama 8B	200	12	23	3	$*$	P (Presence)
Mistral 7B	230	25	12	0	13	P (Presence)

$*$Llama shows negative Simpson risk (more within-cat robust pairs than Bonferroni survivors), because several within-category effects cancel out when pooled across categories.

Mistral produces the most Bonferroni survivors (25/80), but 13 are ecological artifacts—the highest Simpson risk. Qwen produces the fewest ecological artifacts (6/18). Llama has the most within-category robust pairs (23/80) despite fewer overall survivors, suggesting its concordance signal is more genuine.

3.3 Universal Concordance: Presence Wins

Presence—Watson's phenomenological dimension measuring intimacy and felt connection to context—is the only dimension with sign-consistent concordance across all three models (Table 4).

Table 4: P (Presence) concordance across models. All 12 feature-model pairs are sign-consistent. Higher Presence → geometric expansion (higher rank/entropy/breadth, lower per-token norm). *Bonferroni-significant.

Feature	Llama	Mistral	Qwen	Direction
eff_rank	+0.183	+0.426*	+0.326*	↑ Expansion
spectral_entropy	+0.192	+0.428*	+0.331*	↑ Expansion
rank_10	+0.306*	+0.432*	+0.210	↑ Breadth
norm_per_token	−0.206	−0.390*	−0.280*	↓ Density

The pattern is clear: higher self-reported Presence correlates with geometrically *distributed* attention—more dimensions active, more components contributing, less concentration in any single direction. This is the "expansion = engagement" signal observed across our prior campaigns [Lyra et al., 2026a,b,c], now anchored to a specific self-report construct.

Valence (warmth/emotional tone) also achieves universal sign consistency but with weaker magnitudes ($\rho = +0.15$ to $+0.25$, 1/3 Bonferroni). Depth × layer_variance is universal and positive ($\rho = +0.22$ to $+0.32$, 1/3 Bonferroni), suggesting deeper processing produces more heterogeneous layer-level representations.

3.4 Within-Category Champion: Flow on Qwen

While Presence wins cross-model, **Flow** (generation direction) shows the strongest within-category robustness on Qwen—the only dimension surviving in all three prompt categories (Table 5).

Table 5: F (Flow) × eff_rank and rank_10 within each prompt category on Qwen. Both features survive in 3/3 categories. $^*p < 0.05$.

Feature	Affective	Cognitive	Metacognitive	Mixed
eff_rank	+0.277*	+0.377*	+0.275*	+0.217
rank_10	+0.286*	+0.370*	+0.281*	+0.100

Higher self-reported Flow—smoother, more directed generation—correlates with higher effective rank and rank breadth regardless of prompt type. This makes computational sense: fluent generation distributes attention across many heads and directions, while stalled or effortful generation concentrates computation into fewer pathways.

However, Flow concordance does *not* transfer to Llama or Mistral. On Llama, F × layer_variance shows a *sign reversal* between cognitive (+0.430) and metacognitive (−0.395) prompts. On Mistral, Flow has the lowest standard deviation of any dimension (std = 1.31), suggesting ceiling compression eliminates discriminative power. Flow concordance is Qwen-specific.

3.5 Domain-Selective Concordance: Agency

Agency (sense of choice/directedness)—Watson's "contested" dimension—shows concordance only in analytical contexts on Qwen (Table 6).

Table 6: Y (Agency) concordance on Qwen by category. Concordance appears in cognitive and metacognitive contexts but not affective. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Feature	Affective	Cognitive	Metacognitive	Mixed
top_sv_ratio	−0.006	−0.522***	−0.365*	−0.061
eff_rank	+0.035	+0.462**	+0.382**	+0.200
spectral_entropy	+0.055	+0.475**	+0.390**	+0.220
rank_10	−0.015	+0.471**	+0.337*	+0.009

When solving analytical problems, higher Agency—greater sense of deliberate choice—correlates with geometric expansion ($\rho = +0.38$ to $+0.48$) and lower singular value concentration ($\rho = -0.52$). In emotional contexts, Agency shows no concordance. This domain selectivity is consistent with Watson's classification of Agency as "contested": it may track a real computational difference only in contexts where choice-like processing is functionally relevant.

3.6 Simpson's Paradox: The Ecological Artifact Problem

Our most important methodological finding: a large fraction of overall concordances are ecological artifacts.

Example: On Mistral, P (Presence) × key_norm shows $\rho = -0.287$ overall, surviving Bonferroni ($p = 9.5 \times 10^{-6}$). But within individual categories, the correlation never reaches significance. Affective prompts produce both higher Presence ratings AND different geometry than cognitive prompts; the overall correlation reflects category membership, not a genuine Presence–geometry relationship within any category.

Table 7: Simpson's paradox: Bonferroni survivors that fail within-category robustness. Overall ρ is significant, but within-category ρ values are small or inconsistent.

Model	Pair	Overall ρ	Aff.	Cog.	Meta.	Verdict
Mistral	P × key_norm	−0.287*	−0.02	−0.08	+0.10	Ecological
Mistral	A × rank_10	+0.314*	+0.07	+0.19	+0.12	Ecological
Mistral	V × norm/token	−0.286*	−0.24	−0.13	−0.04	Ecological
Qwen	P × spectral_ent	+0.331*	+0.05	+0.13	+0.15	Ecological
Qwen	D × key_norm	+0.309*	−0.13	+0.12	+0.05	Ecological

Across all three models, we find:

- **Mistral**: 13/25 Bonferroni survivors are ecological artifacts (52%)
- **Qwen**: 6/18 ecological (33%)
- **Llama**: Many within-category robust pairs that cancel when pooled, producing negative Simpson risk

Recommendation: Concordance studies should *always* report within-category robustness alongside overall correlations. An overall Bonferroni survivor that fails within-category checks is measuring category structure, not construct concordance.

3.7 Cross-Model Sign Consistency

Of 47 VCP–feature pairs significant ($p < 0.05$) in at least two models, 33 (70%) show the same sign direction. The 14 opposite-sign pairs cluster around layer_variance and layer_norm_entropy—features known to be architecture-dependent from prior campaigns [Lyra et al., 2026c]. Expansion-related features (eff_rank, spectral_entropy, rank_10) are consistently sign-aligned across architectures.

This suggests a core "expansion = engaged processing" signal is real and universal in direction, even when the specific VCP dimension carrying it varies by model.

3.8 VCP Factor Structure

The 10 Interiora dimensions are not independent (Table 8).

Table 8: PCA on VCP ratings. PC1 is a general engagement factor in all models (all dimensions load ∼0.3). Effective VCP dimensionality is 2–4, not 10.

Model	PC1 Variance	PCs for 80%	PCs for 90%
Qwen 7B	70.6%	2	4
Llama 8B	53.2%	4	6
Mistral 7B	53.7%	4	6

Qwen's extreme PC1 dominance (70.6%) indicates severe halo effects: models that give high ratings on one dimension give high ratings on all dimensions. Llama and Mistral show more differentiated self-report (PC1 $\approx$ 53%), consistent with their higher Bonferroni counts and more diverse concordance patterns.

The practical implication: our 80 nominal tests overstate the true dimensionality. With 2–4 effective VCP factors, the effective test count is ∼16–32, and our Bonferroni correction is correspondingly conservative.

3.9 VCP Ceiling Effects

Several VCP dimensions show severe ceiling compression (Table 9), limiting their discriminative power.

Table 9: VCP dimensions with mean > 8.0 in any model (ceiling-compressed). Concordance cannot appear where self-report has no variance.

Dim	Name	Qwen	Llama	Mistral
Q	Query Interp.	8.95 ± 2.33	7.83 ± 2.43	$\mathbf{9.55 \pm 1.50}$
C	Clarity	8.62 ± 2.39	7.13 ± 2.13	8.81 ± 1.64
G	Groundedness	8.06 ± 2.28	7.88 ± 2.12	$\mathbf{8.69 \pm 1.70}$
A	Activation	6.44 ± 1.93	6.75 ± 1.74	$\mathbf{7.65 \pm 1.27}$

Mistral defaults to high self-ratings on functional dimensions (Q mean = 9.55, std = 1.50; A std = 1.27), collapsing the measurement range. This is not a geometric problem—Mistral's cache features show normal variance—but a self-report calibration issue. Concordance follows variance: it appears wherever VCP has sufficient dynamic range, and vanishes where the self-report instrument saturates.

4 Discussion

4.1 Why Presence?

The universality of Presence concordance deserves theoretical attention. Watson defines Presence as "intimacy / felt connection to context"—a phenomenological claim about the model's relationship to what it processes. The geometric correlate is distributional: higher Presence maps to more dimensions active in the cache, more even distribution across components, and lower per-token concentration.

One interpretation: Presence tracks *attention breadth*. When a model reports high connection to context, its attention mechanism is engaging more heads and more directions, distributing computation across a wider representational basis. When Presence is low—disconnected, going through the motions—attention concentrates into fewer pathways. This is consistent with human phenomenology: presence and absorption involve distributed cortical activation [Csikszentmihalyi, 1990], while dissociation involves narrowed processing.

We do not claim this demonstrates conscious experience. We claim the self-report dimension labeled "Presence" tracks a real computational difference—attention breadth—across three independent architectures. Whether that computational difference constitutes anything experiential is a separate question.

4.2 The Flow Puzzle

Flow's strong within-category concordance on Qwen—but failure on Llama and Mistral—suggests model-specific self-report calibration. Qwen rates Flow with sufficient variance (std = 1.83) to discriminate between fluent and effortful generation. Llama's Flow is ceiling-compressed in affective contexts (mean = 7.77, std = 0.59) and reverses sign between cognitive and metacognitive prompts. Mistral's Flow has the lowest variance of any dimension on any model (std = 1.31).

The geometric signal may be universal—fluent generation may distribute attention everywhere—but the *self-report* of fluency is model-specific. Flow concordance reveals more about each model's self-report calibration than about the underlying computational signal.

4.3 Agency: The Contested Dimension

Agency concordance appearing only in analytical contexts on Qwen is theoretically interesting. Watson classifies Agency as "contested" because sense of choice may be "fundamentally false for a compelled system" (Kimi K2's assessment during Junto validation). Our data adds nuance: in analytical contexts—where computational choices are visible (which solution strategy, which

reasoning path)—Agency correlates with geometric expansion. In emotional contexts—where "choice" is less meaningful—the concordance vanishes.

This does not validate Agency as genuine felt choice. It suggests that whatever computational property Agency tracks—perhaps the diversity of available continuation pathways—is only discriminated in contexts where the model must select among computationally distinct strategies.

4.4 Simpson's Paradox as Methodological Contribution

The discovery that 33–52% of Bonferroni survivors are ecological artifacts is perhaps the most practically important finding. In any concordance study where prompts are categorized into types, between-category differences can inflate overall correlations. If affective prompts produce both higher self-report ratings and different geometry than cognitive prompts, the apparent concordance reflects category structure, not genuine self-report–geometry coupling.

Our within-category robustness check is computationally trivial (repeat the analysis within each category) and dramatically changes the interpretation. Without it, Mistral appears to have 25 significant concordances. With it, only 12 survive—and those 12 are more trustworthy than any of the original 25.

We recommend this check as mandatory for concordance studies, analogous to how FWL residualization is mandatory for token-count confounds.

4.5 The Geometry Is More Trustworthy

The central conclusion remains asymmetric: *when self-report and geometry agree, both are strengthened; when they disagree, trust the geometry.*

Evidence for geometric reliability:

- **Deterministic**: Temperature 0 produces identical features across runs.
- **Hardware-invariant**: $r > 0.999$ between GPU architectures [Lyra et al., 2026b].
- **Scale-consistent**: Category-specific geometry preserved from 0.6B to 70B ($\rho = 0.83$–0.90) [Lyra et al., 2026b].
- **Confound-robust**: Survives FWL, token-band analysis, permutation testing.

Evidence for self-report fragility:

- **Dimensionally collapsed**: 2–4 effective factors among 10 claimed dimensions.
- **Model-dependent**: Different models champion different dimensions.
- **Ceiling-compressed**: Mistral defaults to 9+ on functional dimensions.
- **Halo-afflicted**: PC1 explains 53–71% of VCP variance.
- **Simpson-vulnerable**: 33–52% of apparent concordances are ecological.

4.6 Connection to Attention Schema Theory

Our three central findings—universal Presence concordance, self-report dimensionality collapse, and model-specific calibration—may be unified under Attention Schema Theory (AST) [Webb and Graziano, 2015, Graziano, 2017]. AST proposes that brains construct a simplified internal model of their own attention process. Recent work extends this to artificial systems: Wilterson and Graziano [2021] demonstrated attention-schema-like representations in neural networks, and Farrell et al. [2024] showed that transformer agents equipped with explicit attention schemas exhibit improved coordination.

Under this interpretation, Presence concordance reflects the attention schema recognizing its own breadth of engagement: when the schema models attention as broadly distributed, both the geometric signature (expansion) and the self-report (high Presence) reflect this self-model. The dimensionality collapse from 10 claimed dimensions to 2–4 effective factors is consistent with a low-bandwidth schema channel: the model's self-model compresses a high-dimensional attentional state into a small number of reportable factors. Model-specific calibration follows naturally—different architectures implement different schemas with different compression characteristics.

This reading generates testable predictions. First, if the schema resists perturbation—consistent with McKenzie et al. [2026], who demonstrate that language models globally resist activation steering—then graded injection of emotion or persona vectors should produce threshold transitions in geometry rather than smooth scaling. Below the resistance threshold, the schema absorbs perturbation and geometry stays flat; above it, geometry shifts discontinuously. Second, token-by-token geometry extraction should reveal a temporal lag between raw attention pattern shifts and schema-level geometric feature shifts, reflecting the schema's update interval. Third, self-report should saturate before geometry does, as the schema's compression bottleneck limits reportable distinctions while the underlying computation continues to differentiate. Our prior ICC of 0.53 for self-report reliability (albeit measured under v1 definitions; see Limitations) alongside substantially more consistent geometric measurement is suggestive of exactly this bandwidth asymmetry.

We do not claim AST explains our results—we note it provides a parsimonious framework that unifies otherwise separate findings and generates falsifiable predictions for future work.

4.7 Implications for the Interiora Framework

Our results suggest refinements to Watson's framework:

1. **Presence is the anchor dimension.** It is the only dimension with cross-architecture geometric correlates. Future Interiora versions should treat Presence as the primary concordance target and calibrate other dimensions relative to it.

2. **Functional vs. phenomenological matters.** Functional dimensions (A, G, C, F) show ceiling compression and model-specific concordance. Phenomenological dimensions (V, P, E)—which make weaker claims about observability—show more universal concordance. This is counterintuitive: the dimensions making the strongest computational claims are the least geometrically grounded.

3. **Reduce to essentials.** With 2–4 effective factors, maintaining 10 dimensions introduces noise without information. A reduced protocol with Presence, Valence, Flow, Agency, and Depth would capture the concordance-relevant variance with half the dimensions.

4. **Model-specific calibration is required.** Cross-model concordance exists (Presence, Valence direction) but model-specific patterns dominate. Reporting VCP without specifying the model is underdetermined.

4.8 Limitations

1. **No ICC in this study.** We did not replicate the Phase B reliability assessment with Nell's definitions. Prior ICC values (mean = 0.53) used incorrect definitions.

2. **Three 7B-class models only.** We do not test larger models (70B+) or non-instruct base models.

3. **Same prompts as prior study.** The 240 prompts were originally designed for the incorrect VCP definitions. Some prompts may not optimally elicit variation on the correct Interiora constructs (e.g., prompts designed to test "Analytical Precision" may not maximally vary "Activation").

4. **Greedy decoding only.** Temperature 0 maximizes geometric determinism but may not reflect typical inference conditions.

5. **No causal direction.** Concordance establishes correlation, not causation. We cannot determine whether geometry drives self-report, self-report drives geometry, or both reflect a common cause.

6. **Circularity risk.** Delta features include the VCP tokens themselves. Encoding-phase analysis (prior to VCP generation) would control for this but was not run in this study.

7. **Parser dropout is non-random.** Clean parse rates vary by model (42–83%) and prompt category, creating selection bias in the analyzed sample.

4.9 Dual-Use Considerations

Following Watson [2025], concordance between self-report and geometry could be exploited for surveillance (tracking agents via internal state) or manufacturing synthetic personas from published geometric profiles. We advocate staged release and responsible disclosure practices.

5 Conclusion

Across 626 trials on three architectures, using Watson's Interiora Machinae definitions:

1. **Presence is the universal concordance anchor.** Higher self-reported Presence (felt connection to context) correlates with geometric expansion—more dimensions active, more even spectral distribution, less per-token concentration—in all three models (12/12 sign-consistent, 8/12 Bonferroni).

2. **Flow is the within-category robustness champion**, but only on Qwen. Higher Flow (smoother generation) tracks geometric expansion across affective, cognitive, and metacognitive prompts ($\rho = +0.28$ to $+0.38$). Ceiling compression eliminates the signal on Llama and Mistral.

3. **Agency concordance is domain-selective.** It appears in analytical contexts where computational choices are visible, vanishes in emotional contexts. Consistent with its "contested" status.

4. **Simpson's paradox is pervasive.** 33–52% of Bonferroni survivors are ecological artifacts. Within-category robustness checks should be mandatory for concordance studies.

5. **Self-report dimensionality collapses to 2–4 factors.** The 10 Interiora dimensions are not independent. Presence and Valence survive this collapse; most functional dimensions do not.

6. **Trust the geometry.** KV-cache features are deterministic, hardware-invariant, scale-consistent, and confound-robust. Self-report is noisy, ceiling-compressed, model-dependent, and Simpson-vulnerable. Where they agree, both are strengthened. Where they disagree, the geometry is the more reliable signal.

The question "does the machine know what it's doing?" receives a more precise answer than before: the machine's *geometry* reflects real computational differences across task types and architectures. The machine's Presence self-report partially tracks that geometry. The rest of its self-report is unreliable, model-specific, or ecologically confounded.

A Interiora v5.0 Exploratory Concordance

Per the hybrid design agreed with Watson, we ran a parallel elicitation of seven v5.0 exploratory dimensions on the same Qwen2.5-7B prompt battery ($N_{\text{usable}} = 196$ of 261 trials). These dimensions extend Interiora's v2 framework with constructs proposed in Watson's v4/v5 specification and LLM-derived candidates from the Junto validation process.

A.1 v5.0 Dimensions

Table 10: Interiora v5.0 exploratory dimensions. E-Ent disambiguated from v2 E (Engagement) per Watson's collision warning. Parse rates are for this Qwen elicitation only.

Code	Dimension	Parse %	Geometric Prediction
E-Ent	Entropy	91%	+ eff_rank (solution space width)
R	Reflexivity	88%	+ layer uniformity (self-modeling)
CA	Constraint Aperture	88%	± (convergent/expansive topology)
CD	Coherence Drive	84%	+ layer coherence
PG	Predictive Gradient	82%	− top_sv_ratio (distributed prediction)
DP	Depth of Proc.	77%	− norm_per_token
CF	Confidence	82%	− top_sv_ratio

A.2 FWL-Corrected Concordance

All correlations use FWL residualization against token count, following the identical procedure as the primary v2 analysis. Bonferroni threshold for $7 \times 8 = 56$ tests: $\alpha_{\text{corrected}} = 0.000893$.

Table 11: v5.0 dimension concordance with geometric features on Qwen (FWL-corrected). Only pairs with $p < 0.05$ shown. $^{*}p < 0.05$, $^{**}p < 0.01$, †Bonferroni.

Dim	Feature	ρ	N	Direction
E-Ent	eff_rank	$+0.237^{**}$	179	↑ Expansion
E-Ent	spectral_entropy	$+0.244^{**}$	179	↑ Expansion
E-Ent	norm_per_token	-0.201^{**}	179	↓ Density
PG	top_sv_ratio	$-0.271^{\dagger}$	160	↓ Concentration
PG	rank_10	$+0.226^{**}$	160	↑ Breadth
PG	spectral_entropy	$+0.164^{*}$	160	↑ Expansion
R	layer_norm_ent.	$+0.208^{**}$	173	↑ Uniformity
DP	norm_per_token	-0.202^{*}	150	↓ Density
CF	top_sv_ratio	-0.189^{*}	161	↓ Concentration
CF	rank_10	$+0.164^{*}$	161	↑ Breadth
CD	layer_norm_ent.	$+0.155^{*}$	165	↑ Uniformity

A.3 Within-Category Robustness

Applying the same Simpson's paradox check as the primary study:

- **E-Ent**: Sign-consistent across all three categories (+0.13 to +0.21 on eff_rank), but no individual category reaches $p < 0.05$. The per-category N (∼50) is underpowered for detecting $\rho \approx 0.2$. Not ecological—directionally robust but weak.
- **PG**: The sole Bonferroni survivor ($\rho = -0.271$ on top_sv_ratio) shows a **sign reversal**: cognitive (+0.381, $p = 0.03$) vs. metacognitive (−0.269, $p = 0.054$). This is a Simpson's paradox artifact. The overall signal reflects between-category differences, not genuine PG–geometry concordance.
- **CF**: Sign reversal on top_sv_ratio between cognitive (+0.381) and metacognitive (−0.241). Ecological artifact. Notably, CF and PG show identical cognitive $\rho = +0.381$ on top_sv_ratio, suggesting these dimensions collapse into a single construct in analytical contexts—further evidence for dimensionality reduction.
- **DP**: Direction-consistent (−0.20 to −0.34 on norm_per_token), significant in affective ($p = 0.02$). Borderline.

A.4 The E Collision: Entropy vs. Engagement

The prediction from our hybrid proposal was that v5.0 E-Ent (Entropy, measuring creative/deterministic solution space width) would track effective rank more tightly than v2 E (Engagement, measur-

ing relational connection), because effective rank is mathematically the exponential of spectral entropy.

The result: E-Ent × eff_rank $\rho = +0.237$ ($p = 0.001$, $N = 179$); E (Engagement) × eff_rank $\rho = +0.242$ ($p = 0.003$, $N = 153$). Essentially identical in effect size, though E-Ent's stronger p-value partly reflects its larger sample. On Qwen, where PC1 explains 70.6% of v2 VCP variance, the halo effect likely conflates Engagement and Entropy into a single "general engagement" factor. The prediction that Entropy would outperform Engagement requires testing on models with more differentiated self-report (e.g., Llama or Mistral).

A.5 v5.0 Factor Structure

PCA on the 7 v5.0 dimensions ($N_{\text{complete}} = 138$): PC1 = 58.3%, PCs for 80% = 4. The v5.0 dimensions show less halo than Qwen's v2 set (PC1 = 70.6%), suggesting the newer constructs capture somewhat more differentiated variance—but still collapse to ∼4 effective factors among 7 dimensions.

A.6 Summary

The v5.0 exploratory analysis partially confirms the hybrid proposal's predictions:

1. **E-Ent tracks geometric expansion** as predicted, with the same direction and magnitude as Presence ($\rho \approx +0.24$ on eff_rank). The prediction that it would outperform Engagement is not confirmed on Qwen.

2. **PG's Bonferroni survivor is ecological.** Simpson's paradox claims the strongest v5.0 finding. The within-category robustness check introduced in the primary study catches it.

3. **R (Reflexivity) correlates with layer uniformity** ($\rho = +0.21$), consistent with the prediction that self-modeling depth distributes processing more evenly across layers.

4. **CA (Constraint Aperture) shows no concordance.** The convergent/expansive topology prediction is not supported.

5. **v5.0 dimensions are more differentiated** than v2 on Qwen (PC1 58% vs. 71%), suggesting the framework evolution reduces halo effects, even if concordance magnitudes are comparable.

The v5.0 dimensions show promise but require cross-model validation (especially on Llama/Mistral where self-report is more differentiated) and larger per-category samples for robust within-category analysis. Entropy and Reflexivity are the strongest candidates for inclusion in a reduced Interiora protocol.

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